CS342 · VISUAL PROGRAMMING

Midterm 471 Solutions

Find Output and writing a GUI

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01 Solution Q1 (a): tab2

Each element is one iteration, and the catch blocks are inside the for, so an exception ends only that iteration and the loop continues.

i	list[i]	What happens	Prints	sum · nbNotes
0	"-1"	Negative → ExceptionRange("few")	Note no.1 is too few	0 · 0
1	"2"	Between 0 and 20 → added	—	2 · 1
2	"x"	**NumberFormatException**	Note no.3 is not integr	2 · 1
3	"30"	Above 20 → ExceptionRange("large")	Note no.4 is too large	2 · 1
4	"18"	Added	—	20 · 2

nbNotes = 2 is not zero, so it returns `20 ÷ 2 = 10` and main prints it.

```
Note no.1 is too few
Note no.3 is not integr
Note no.4 is too large
The average is: 10
```

(i+1) because counting starts at 0. getMessage() returns the text passed to super(s): few or large.

02 Solution Q1 (b): tab3

i	list[i]	Why it is not an integer	Prints
0	"x"	A letter	Note no.1 is not integr
1	" "	A space	Note no.2 is not integr
2	"2.3"	Has a decimal point; parseInt accepts whole numbers only	Note no.3 is not integr
3	"/n"	Plain text (not a new line)	Note no.4 is not integr

After the loop nbNotes = 0, so the method throws ExceptionEmpty. main catches it and prints er; println(er) prints what toString() returns.

```
Note no.1 is not integr
Note no.2 is not integr
Note no.3 is not integr
Note no.4 is not integr
No note is valid
```

"The average is: " is never printed, because the exception happens while evaluating average(tab3), before println completes.

The code spells it "integr", and the output matches it letter for letter.

03 Solution Q2

Line	Real type	What happens	Prints
((Phone) d1).call()	Phone	A valid cast; call exists in Phone	Phone is calling
d2.start()	Laptop	No cast, but the override runs by the real type (polymorphism)	Laptop is starting
((Phone) d1).start()	Phone	Phone's version	Phone is starting
((Laptop) d2).close()	Laptop	Laptop does not override close, so the parent's version runs	Device is closing
((Phone) d3).call()	Devices	The object is the parent, not a Phone → ClassCastException	The program stops

```
Phone is calling
Laptop is starting
Phone is starting
Device is closing
Exception in thread "main" java.lang.ClassCastException
```

The last line compiles (the compiler trusts you), but at run time it fails because the real object new Devices() is not a Phone. This is the downcasting rule from Chapter 1.

Note: Phone and Laptop are written inside Devices. As non-static inner classes, new Phone() inside the static main would not compile. The solution above treats them as ordinary classes, which is what the question intends.

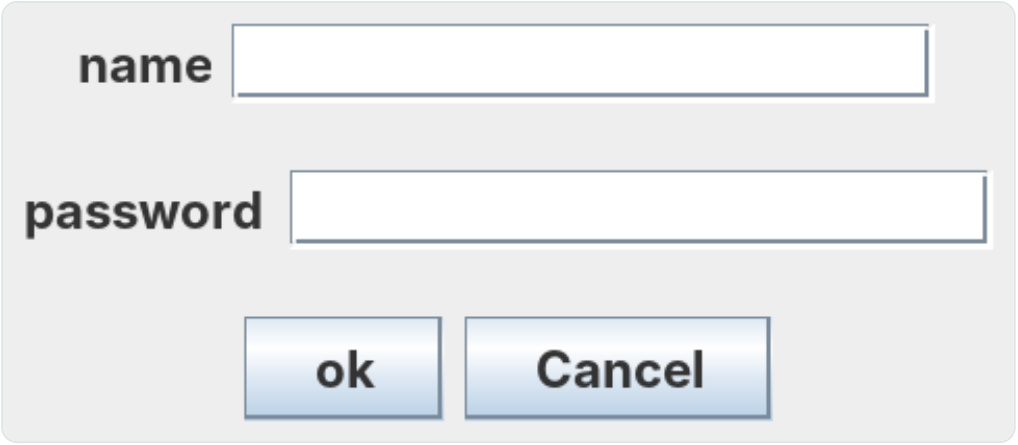
04 Solution Q3

The window has 3 rows, so the content pane takes GridLayout(3, 1) and each row is a JPanel

with a FlowLayout. Row 2 is LEFT because password starts at the left edge.

```
import javax.swing.*;
import java.awt.*;
public class Q3 extends JFrame {
    JLabel l1, l2;
    JTextField t1;
    JPasswordField t2;              // hides the characters
    JButton b1, b2;
    public Q3() {
        setTitle("login");
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        Container p = getContentPane();
        p.setLayout(new GridLayout(3, 1));
        JPanel p1 = new JPanel(new FlowLayout(FlowLayout.CENTER));
        JPanel p2 = new JPanel(new FlowLayout(FlowLayout.LEFT));
        JPanel p3 = new JPanel(new FlowLayout(FlowLayout.CENTER));
        l1 = new JLabel("name");
        t1 = new JTextField(15);
        l2 = new JLabel("password");
        t2 = new JPasswordField(15);
        b1 = new JButton("ok");
        b2 = new JButton("Cancel");
        p1.add(l1);  p1.add(t1);
        p2.add(l2);  p2.add(t2);
        p3.add(b1);  p3.add(b2);
        p.add(p1);
        p.add(p2);
        p.add(p3);
        pack();                      // smallest size that shows everything
        setVisible(true);
    }
    public static void main(String[] args) {
        new Q3();
```

```
    }
}
```


name
password
ok
Cancel